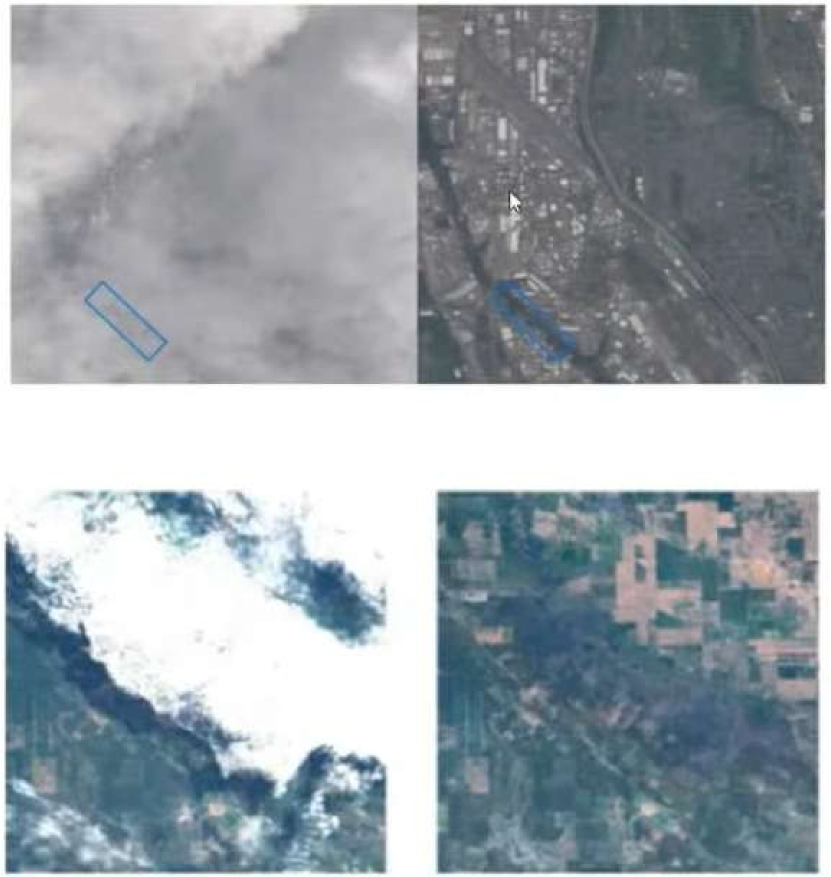
Introduction

Cloud Removal

- Image to image translation
- Dense relation mapping
- Image synthesis
- Temporal (Single, Multi frame)
- Modality (Single, Multi-modal)



Visualization of Cloud removal

Source: Li et al. 2024

Introduction

Need

- Persistent cloud cover
- Image synthesis for gap filling

Visualization of Cloud removal

Objectives

- Develop a Generative AI-based framework for automated cloud removal in LISS-IV imagery.
- Reconstruct cloud-covered regions while preserving spatial structures and spectral characteristics.
- Generate visually consistent and analysis-ready cloud-free imagery.
- Evaluate the reconstructed outputs using quantitative and qualitative assessment methods.
- Develop a scalable workflow for operational cloud removal applications.

Expected Outcomes

- Automated cloud-free reconstruction of LISS-IV imagery.
- Improved usability of optical satellite data under persistent cloud-cover conditions.
- Enhanced spatial and spectral consistency in reconstructed outputs.
- Generation of analysis-ready satellite products for downstream geospatial applications.
- Development of a prototype framework for operational deployment.
- Comparative assessment of different Generative AI architectures for cloud reconstruction.

Dataset Required

- Primary Dataset
  - LISS-IV satellite imagery (cloudy and cloud-free scenes) (Bhoonidhi)
- Auxiliary Datasets (Optional publicly available data)
  - Sentinel-1 SAR imagery
  - Sentinel-2 optical imagery
  - Temporal reference imagery
  - DEM data
  - Any other datasets

Suggested Tools/Technologies

- Programming Frameworks
  - Python
  - PyTorch / TensorFlow
- Geospatial Tools
  - GDAL
  - Rasterio
  - QGIS
  - Google Earth Engine (optional)
- Supporting Libraries
  - OpenCV , NumPy, Scikit-image, Albumentations

Steps

- **Collection and preprocessing of cloudy and cloud-free LISS-IV imagery.**
- Preparation of cloud masks and co-registration of auxiliary datasets (optional).
- Identification and reuse of suitable pre-trained deep learning or Generative AI models for transfer learning (optional).
- Fine-tuning of pre-trained models on LISS-IV imagery for cloud reconstruction tasks (optional).
- Integration of temporal and/or multi-sensor information for improved reconstruction quality (optional).
- **Training, optimization, and validation of the developed deep learning model.**
- **Generation of analysis-ready and cloud-free products.**
- **Documentation and dep** [illegible] **oped workflow.**

Evaluation

- Visual consistency of reconstructed regions
- Preservation of spatial textures and edges
- Expert visual assessment